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SURVMETH/SURV 625: Applied Sampling - Project Report

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1. INTRODUCTION

The State of Michigan Department of Education has tasked us with creating a sample design for a survey regarding present teens (13–19 years of age) smoking and drug use habits. The department has requested the design of a statewide sample of teenagers that will help officials monitor the overall rates of smoking and drug use, and the regional variations within the state. This will provide insights into how these behaviors differ across different communities, regions, and demographic groups within Michigan. As sample design experts, our task is to develop a sound and cost-effective survey design that meets the state's precision requirements. This includes determining appropriate sample sizes, selecting the first-stage units, and ensuring that the sampling procedure provides reliable estimates. We must also define the methods for stratifying the first-stage units, allocating the sample across different strata, and selecting samples in a way that provides an equal probability of selection.

The state has opted for a school-based sampling approach. Despite the coverage challenges of a school-based survey, the state believes that this method is the right balance between accuracy, cost, and logistical feasibility. The population for this study will consist of all public and private school students in grades 7 through 12 across Michigan. The survey is scheduled for the upcoming fall semester, and the collected data will inform health interventions, policy decisions, and provide a baseline for future monitoring. It is important to note that potential errors in nonresponse, noncoverage, and mode of administration will be accepted due to prior decisions by state officials.

To ensure the design is comprehensive and effective, the team will rely on additional data and expert consultation. Dr. Si, a consultant in sampling methodology, will provide guidance as needed. As part of the project, we will collaborate as a team of sampling experts to develop a final sampling plan and deliver an actionable report.

2. SAMPLE SIZE DETERMINATION

The analysis reveals striking differences in required sample sizes across variables, driven by their inherent variability and prevalence.

We see that Ever Used Marijuana stands out, demanding the largest sample (2,267) to achieve a 5% coefficient of variation. This reflects its lower prevalence (15%) and higher relative variability, which amplifies uncertainty and necessitates more observations for precise estimation.

Ever Smoked, with a higher prevalence (25%), requires fewer samples (1,200). The broader base of individuals who've smoked reduces variability, making it easier to estimate reliably.

Most notably, Age First Approached defies expectations that only 3 samples are needed. This suggests remarkably low variability in the population (variance = 1.0), with most individuals reporting their first exposure around age 12. Interestingly, the tightly clustered data means even a tiny sample accurately captures the mean, highlighting a well-defined behavioral pattern.

3. REGIONAL STRATIFICATION ANALYSIS

Now, the analysis of Michigan school data provides a clear snapshot of how student populations are distributed across different regions and how this impacts sample allocation for future studies. By mapping each county to its respective region and then summing up the total number of students and schools per region, we get a detailed view of the educational landscape.

The results show that Region 9 stands out by far, containing nearly half (45%) of the state's student population—over 374,000 students across 923 schools. Regions 7 and 8 also have substantial student bodies, each representing about 23% of the total. In contrast, regions like 1, 2, 3, and 4 have much smaller populations, each accounting for less than 1% of the total students.

When it comes to proportionate sample allocation, these differences translate directly into how many students should be sampled from each region to ensure fair and representative coverage. For example, if one plans on sampling 4,224 students statewide, almost 1,900 would come from Region 9 alone, while only 18 would be sampled from Region 1.

This approach ensures that larger, more populous regions are appropriately represented in any statewide analysis, while smaller regions are not overlooked. It's a practical application of proportional allocation in survey sampling, ensuring that the diversity and scale of Michigan's student population are accurately reflected in any research or policy decisions.

4. OPTIMAL SAMPLING DESIGN

The results highlight how intra-cluster correlation (ICC) directly shapes sampling efficiency. With ICC values ranging from 0.014 (Age First Asked) to 0.031 (Ever Smoked), schools exhibit moderate homogeneity in student behaviors—particularly for smoking.

This clustering effect amplifies the required sample size, as seen in the design effect of 2.32, meaning nearly twice as many students are needed compared to a simple random sample. Weeks 7 and 8 gave us the cluster and sample size, which, balancing cost and precision, the optimal design selects 96 schools with 44 students each (totaling 4,224 students). This strikes a middle ground between two extremes: surveying 166 schools (maximizing geographic spread but undersampling students) or 10,000 students (ignoring school-level costs entirely). The choice reflects real-world constraints like school recruitment expenses ($2,500 per school) and per-student data collection costs ($100), while prioritizing statistical power.

- Higher ICCs (e.g., smoking behavior) demand more clusters to capture between-school variability.
- The 44-student cluster size minimizes variance inflation from within-school similarities.
- Budget allocation mirrors practices in education research, where fixed cluster costs often dominate.

Practical considerations:

- Non-response adjustments: building a 10-15% buffer to ensure final samples meet power requirements.
- Parameter justification: benchmarking against prior studies (e.g., similar school-based smoking surveys) strengthens methodological credibility.
- Cost transparency: explicitly separating cluster/student costs helps stakeholders understand tradeoffs.

5. OPTIMUM SUBSAMPLE SIZE

Variable	**Rho**	**DEFF**	**SRS.Variance**	**Cluster.Sample.Variance**	**Precision**
Smoked	0.0306	2.32	4.44E-05	0.000103	0.0101
Marijuana	0.0204	1.88	3.03E-05	5.68E-05	0.00754
Age (First Use)	0.0143	1.61	0.000236	0.000382	0.0195

Table 1. Table showing variables and their respective values.

In Week 7, we refined our cluster sampling design to optimize both cost efficiency and statistical precision. Using the intraclass correlation coefficients (ICCs) for the three key variables—ever smoked (0.0306), ever used marijuana (0.0204), and age at first use (0.0143)—we calculated the optimal number of students to sample within each cluster. We used the following equation:

$$m_{opt} = \sqrt{\frac{C_n}{C_m} \cdot \frac{1-p}{p}}$$

Our C_n being 3000 and C_m being 50, this resulted in recommended subsample sizes of 44, 54, and 65 students, respectively. These figures reflect a balance between the increased precision from larger subsamples and the rising cost of sampling additional students.

Additionally, using the following equations:

$$\text{Sampling Cost} = n \cdot c_n + n \cdot m \cdot c_m$$

Given the recommended subsample sizes, we were able to compute the recommended number of clusters for each subsample size which resulted in 96, 88, and 80, respectively.

Given our $500,000 budget, we found that sampling 44 students per cluster would allow for the inclusion of approximately 96 clusters, resulting in a total sample size of around 4,224 students. This configuration stays within budget while maintaining a manageable cluster size.

To quantify the impact of clustering, we used the design effect formula:

$$var_{cluster-new}(\bar{y}) = var_{srs} \cdot deff_{new}$$

This yielded design effects of 2.32 for smoking, 1.88 for marijuana use, and 1.61 for age at first use, indicating the degree to which variance is inflated due to clustering.

We then calculated the variances under both simple random sampling (SRS) and the proposed cluster design. This demonstrated the SRS variance for smoking was approximately 0.000044, which increased to 0.000103 under the cluster design, as expected due to the larger design effect.

The sampling fraction, calculated as the ratio of the total sample size to the population size (4,231 / 830,138), was approximately 0.0051. This small fraction corresponds to a finite population correction (FPC) factor of about 0.9949, indicating insignificant impact on the variance estimates.

To evaluate precision, we computed the standard errors (square roots of the cluster variances). The resulting values were approximately 0.0101 for smoking, 0.0075 for marijuana use, and 0.0195 for age at first use—suggesting the design achieves a reasonable level of statistical accuracy.

The design effects across variables—each close to 1—indicate that clustering does not substantially inflate variance compared to a simple random sample, when selecting a subsample size of 44. Additionally, all cluster sample variances were below 0.01, further supporting the statistical efficiency of the design. Since the FPC is close to 1, its influence is minimal, and thus we concluded it could be ignored. Based on these findings, we felt confident in choosing the smallest optimal subsample size (44 students) as sufficient to maintain precision while minimizing costs.

6. ALLOCATED CLUSTERS AND PPES

Table 2. Sampling Parameters by Region

Region	Total Students	Number of Schools	Proportion	Allocated Clusters	1st-Stage Sampling Fraction	Zone Size	Random Start	1st-Stage Fraction (Stratum)
1	2893	16	0.0035	1	0.004	178	142	0.043
2	5474	30	0.0066	1	0.007	182	137	0.028
3	8044	30	0.0097	1	0.010	262	102	0.014
4	4855	31	0.0058	1	0.006	157	53	0.012
5	14692	69	0.0177	2	0.037	236	85	0.009
6	31140	123	0.0375	4	0.156	249	49	0.006
7	177421	598	0.2137	21	4.671	298	159	0.018
8	178447	517	0.2150	21	4.698	344	33	0.004
9	374755	923	0.4514	44	20.670	406	401	0.047

This statewide education study aimed to balance efficiency and fairness by dividing Michigan into nine geographic regions and treating each as its own unique "stratum." Regions with more students—like Region 9, which had nearly half the state's student population—naturally got more schools. The table above demonstrated that Region 9 received 44 of the 96 total clusters. Regions with fewer students—like Region 1, received 1 of the 96 total clusters as it has the smallest student population. This proportional allocation ensured heavily populated areas weren't overlooked, while smaller regions still had a voice.

Next, we calculated the first stage sampling fraction for each stratum by multiplying the allocated clusters by the total number of students and dividing by the sum of the total number of students to determine how many schools were going to be selected within each region. Then we began the systematic selecting of schools. We computed implicit stratification by arranging each stratum by the total number of students in ascending order. This allowed our regions to have a spread out sample and we were no longer randomly selecting schools. We computed the zone size for each school using the following equation:

$$\frac{MOS}{\text{region count}}$$

For Region 9 we had a zone size of 406, while for Region 1 the zone size was 178. A random starting point was selected; it would be between 0 and our zone size for each stratum. For example, for Region 9 the random start was 401 and for Region 1 the random start was 142. By using our random start, allocated clusters, and zone size we were able to select our final sample. We chose the amount of schools that were determined by the allocated clusters for each region.

Finally, we checked the math. For each region, we calculated what percentage of its students would end up in the sample based on how many schools we selected and their average size. This "First-stage sampling fraction within each stratum" acted like a reality check, confirming that our design didn't accidentally over- or under-represent any area. The results were as demonstrated in the table found above. By blending stratification, proportional allocation, and size-aware sampling, this method ensured the study was both statistically robust and practical—similar to tailoring a suit to fit Michigan's educational landscape perfectly.

7. STRATIFIED PPES SAMPLING DESIGN AND IMPLEMENTATION

In Week 10, we successfully implemented a stratified Probability Proportional to Size (PPeS) sampling design to maintain EPSEM (Equal Probability of Selection Method) across all strata. The required subsample sizes for each region were computed based on the overall sampling fraction and allocated clusters. Schools that did not meet the required subsample size were combined into linked units to ensure proportional representation while preserving the integrity of the design. Systematic sampling was applied within each school to select students at the specified second-stage sampling rate, ensuring efficiency and reproducibility. The overall probability of inclusion for each student was calculated, confirming that the design achieved EPSEM. This process highlights the robustness of the stratified PPeS design in balancing proportional representation and equal probability of selection across diverse strata.

TASK 1: COMPUTE REQUIRED SUBSAMPLE SIZES

Given the overall m_{opt}, n_{opt}, and N (based on the sampling frame), the overall sampling fraction f is computed as:

$$f = \frac{n_{opt}}{N}$$

For each of the nine strata, the required number of students to subsample from each sampled school is calculated using:

$$\text{Required Subsample Size} = \frac{\text{Total Students} \times f}{\text{Allocated Clusters}}$$

TASK 2: HANDLING SCHOOLS WITH INSUFFICIENT SIZE

If schools in a region do not meet the required subsample size, they are combined into a single unit. This ensures that the required subsample size is met while maintaining proportional representation.

TASK 3: SYSTEMATIC SAMPLING

To physically select the subsample of students within a sampled school:

- Calculate the second-stage sampling rate:

$$f_{2nd} = \frac{m_h}{\text{MOS}}$$

- Perform systematic sampling using the interval:

$$k = \frac{\text{Total Students in School}}{\text{Required Subsample Size}}$$

- Select students at regular intervals starting from a random start.

TASK 4: OVERALL SAMPLING FRACTION AND PROBABILITY OF INCLUSION

The overall probability of inclusion for a student is:

$$\pi_{hi} = \pi_h^{(1)} \cdot \pi_{hi}^{(2)} = \frac{m_{hi}}{\sum \text{MOS}_h}$$

The overall sampling fraction is:

$$f_h = \frac{n_h}{N_h}$$

RESULTS

Region	Total Students	First-Stage Prob	Second-Stage Prob	Overall Prob	Sampling Fraction
1	2893	0.003627	0.005185	1.9×10^{-5}	0.005185
2	5474	0.006862	0.005115	3.5×10^{-5}	0.005115
3	8044	0.010084	0.005097	5.1×10^{-5}	0.005097
4	4855	0.006086	0.005149	3.1×10^{-5}	0.005149
5	14692	0.018417	0.002518	4.6×10^{-5}	0.002518
6	31140	0.039036	0.001252	4.9×10^{-5}	0.001252
7	177421	0.222410	0.000242	5.4×10^{-5}	0.000242
8	178447	0.223696	0.000241	5.4×10^{-5}	0.000241
9	374755	0.469782	0.000115	5.4×10^{-5}	0.000115

WITHIN-SCHOOL SAMPLING AND EPSEM VERIFICATION

In Week 11, we focused on implementing within-school sampling for a randomly sampled middle school in Region 7, ensuring that the design maintains EPSEM (Equal Probability of Selection Method) in a two-stage PPeS sampling framework. The following key observations and results were derived:

ACTUAL SCHOOL SIZE

The actual size of the sampled middle school, based on the classroom rosters, was determined to be:

$$M_{actual} = 219$$

This value differed slightly from the Measure of Size (MOS) of 242 used in the sampling frame. This discrepancy highlights the importance of verifying the actual school size during the sampling process to ensure accurate implementation of the design.

WITHIN-SCHOOL SAMPLING RATE

To achieve EPSEM, the second-stage sampling rate (f_{2nd}) was calculated as:

$$f_{2nd} = \frac{m_7^*}{\text{MOS}}$$

Substituting the values for Region 7:

$$f_{2nd} = \frac{3}{242} = 0.0124$$

This rate ensures that the overall probability of inclusion for each student remains consistent across the stratum.

SAMPLING INTERVAL

The sampling interval (k) was calculated as:

$$k = \frac{1}{f_{2nd}} = 80.67$$

This interval was used to systematically select students from the roster.

EXPECTED SAMPLE SIZE

The expected sample size was calculated as:

$$\text{Expected Sample Size} = M_{actual} \times f_{2nd}$$

Substituting the values:

$$\text{Expected Sample Size} = 219 \times 0.0124 = 38.59$$

This value was rounded to 39 students for practical implementation.

SYSTEMATIC SAMPLING IMPLEMENTATION

Systematic sampling was applied to select students from the roster:

- A random start was selected between 1 and the sampling interval ($k = 80.67$).
- Students were selected at regular intervals (k) from the roster.
- The actual sample size obtained was 39 students, matching the expected sample size.

VERIFICATION OF EPSEM

To verify EPSEM, the following probabilities were calculated:

- **First-Stage Probability**:

$$\pi_h^{(1)} = \frac{\text{MOS}_{hi}}{\sum \text{MOS}_h}$$

- **Second-Stage Probability**:

$$\pi_{hi}^{(2)} = \frac{\text{Actual Sample Size}}{\text{Actual School Size}}$$

  Substituting the values:

$$\pi_{hi}^{(2)} = \frac{39}{219} = 0.1781$$

- **Overall Selection Probability**:

$$\pi_{hi} = \pi_h^{(1)} \cdot \pi_{hi}^{(2)}$$

  The overall selection probability was verified to be consistent with the intended value of:

$$\pi_{hi} = \frac{1}{m_7^*} = \frac{1}{3} = 0.3333$$

COMPARISON WITH POPULATION DISTRIBUTION

The selected sample was compared with the population distribution by grade. The results confirmed that the sample was representative of the population, with minimal deviations in the percentage distribution across grades.

EFFECTIVE SAMPLING RATE

The effective sampling rate was calculated as:

$$\text{Effective Sampling Rate} = \frac{\text{Actual Sample Size}}{\text{Actual School Size}} = \frac{39}{219} = 0.1781$$

This rate aligns with the intended within-school sampling rate, confirming the accuracy of the sampling process.

KEY RESULTS

CONCLUSION

The within-school sampling process successfully maintained EPSEM by ensuring equal probability of selection for all students. The systematic sampling approach was efficient and reproducible, and the results aligned with the expected values. This demonstrates the robustness of the stratified PPeS design in achieving proportional representation and maintaining EPSEM across all strata.

Table 4. Verification of EPSEM Property

Parameter	**Value**
School MOS	242.000000
Actual School Size	219.000000
Sampling Interval (m_7^*)	3.000000
Intended Within-School Sampling Rate	0.0124
Sampling Interval (k)	80.67
Expected Sample Size	38.59
Actual Sample Size	39
Effective Sampling Rate	0.1781
Overall Selection Probability ($1/m_7^*$)	0.3333

8. WEEK 13

SAMPLE DESIGN SPECIFICATION

Appendix Table 2 shows the number of Sampling Error Computation Units (SECUs) per stratum. Regions 1–4 each contain only one SECU, posing challenges for variance estimation, which typically requires at least two units per stratum. These strata may need to be collapsed, potentially affecting variance estimates and degrees of freedom. Regions 5 and 6 have slightly more SECUs (2 and 4), but still may yield unstable variance estimates. In contrast, Regions 7–9 have 21, 21, and 44 SECUs respectively, supporting more reliable variance estimates.

The `tot_all` column, representing the total number of students per school, is used as the size measure (`SECU_Size`). The wide variation in SECU sizes indicates unequal selection probabilities, suggesting either a probability proportional to size sampling approach or differing sampling intensities across SECUs.

VARIANCE ESTIMATION FRAMEWORK

Task 2 outlines variance estimation for the stratified cluster design established in Task 1. Standard methods like Taylor Series Linearization are appropriate for estimating variance of complex statistics, such as means and proportions. The overall variance is calculated by summing within-stratum estimates (see Formulas 2 and 3). Confidence intervals use the point estimate, standard error, and a t-distribution critical value (Formula 4).

Degrees of freedom are estimated as the number of Sampling Error Computation Units minus the number of strata:

$$\mathrm{df} = (96 - 9) = 87 \tag{1}$$

However, since some strata contain only one SECU, collapsing these for variance estimation will lower the effective degrees of freedom.

A suitable example estimator is the weighted stratified mean (Formula 6). Given the variation in school size (`tot_all`), weights are essential to correct for unequal selection probabilities. Proper weighting ensures unbiased estimates and reflects the underlying sample design, whether due to probability proportional to size or varying sampling rates.

SUBCLASS ANALYSIS CONSIDERATIONS

Task 3 focuses on estimating and making inferences for a population subclass comprising about 20% of the total population. Confidence interval construction for this subclass (e.g., $\hat{\theta}_{\mathrm{sub}}$) uses the same variance estimation methods as in Task 2—primarily Taylor Series Linearization—restricted to sample units within the subclass. The same degrees of freedom (87) are typically applied in the t-distribution.

However, the adequacy of the Sampling Error Computation Unit structure for reliable subclass analysis is a concern. While the full sample supports population estimates, analyzing a smaller group reduces the effective sample size. Many SECUs, especially those with small enrollments, may contribute few or no subclass members, leading to unstable variance estimates and inflated standard errors. This sparsity reduces the precision of subclass estimates, and the nominal degrees of freedom may overstate actual reliability.

FORMULAS

$$\mathrm{Var}(\hat{Y}) = \sum_{h=1}^{H} \frac{n_h}{n_h - 1} \sum_{i=1}^{n_h} (y_{hi} - \bar{y}_h)^2 \tag{2}$$

$$\mathrm{Var}_h = \frac{1}{n_h(n_h - 1)} \sum_{i=1}^{n_h} (y_{hi} - \bar{y}_h)^2 \tag{3}$$

$$\mathrm{CI}(\hat{\theta}) = \hat{\theta} \pm t_{\mathrm{df},1-\alpha/2} \cdot \sqrt{\hat{V}(\hat{\theta})} \tag{4}$$

$$\mathrm{df} = \text{Total SECUs} - \text{Total Strata} = 96 - 9 = 87 \tag{5}$$

$$\hat{Y}_w = \sum_{h=1}^{H} \sum_{i=1}^{n_h} w_{hi} y_{hi} \tag{6}$$

where:

- H: Total number of strata
- n_h: Number of sampled SECUs in stratum h
- y_{hi}: Weighted value for SECU i in stratum h
- $\bar{y}_h$: Mean of y_{hi} in stratum h
- w_{hi}: Final sampling weight for SECU i in stratum h
- $\hat{V}(\hat{\theta})$: Estimated variance of estimate $\hat{\theta}$
- $t_{\mathrm{df},1-\alpha/2}$: Critical value from t-distribution